A Hybrid Approach to Rooftop Type Detection in High-Resolution Satellite Images

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Abstract

Rooftop classification, which plays a critical role in urban energy efficiency analysis, can be automated from high-resolution satellite images. Aiming to achieve high accuracy in this task, this study introduces an innovative hybrid model based on the DFC23 Track1 dataset. Our work adopts a two-stage approach that decouples the object detection and classification tasks, utilizing MMRotate for precise localization followed by a YOLOv11-based model for classification. The proposed hybrid model demonstrates superior performance, achieving a general F1-score of 85% on the baseline dataset. The model's adaptability was further proven as its F1-score increased to 89% after fine-tuning on the target Shanghai dataset. The results reveal that this modular and decoupled architecture is an effective alternative for achieving high accuracy and is highly adaptable to different geographical regions.

1. Introduction

In the domain of computer vision, deep learning has achieved state-of-the-art performance on object detection tasks, a success largely fueled by the advent of architectures such as Convolutional Neural Networks (CNNs) and Transformers. These breakthroughs have enabled robust, high-precision object detection, semantic segmentation, and classification across a wide array of application domains, including medical image analysis, autonomous systems, and automated surveillance. Notably, models like YOLO (You Only Look Once) have established a pivotal benchmark for real-time detection by optimizing the trade-off between inference speed and accuracy, and have been successfully deployed in applications ranging from pedestrian detection to industrial automation [1, 2].

Concurrently, in remote sensing, advancements in aerial and satellite imaging technologies—particularly higher spatial resolution, multispectral sensors, and UAV-based (Unmanned Aerial Vehicle) data acquisition—have significantly expanded the scope of deep learning. This has extended its utility to large-scale environmental monitoring, urban planning, and disaster management [3]. Aerial and satellite imagery yields crucial geospatial data that characterizes terrestrial features like buildings and land cover, finding critical applications in cartography, climate change analysis, and urban development [4]. The increasing fidelity of high-resolution remote sensing imagery, in conjunction with sophisticated deep learning models, now facilitates the fine-grained detection and classification of small-scale objects, such as rooftops. The accurate classification of rooftop types is a task of growing importance across multiple disciplines. Consequently, rooftop typology classification, leveraging a spectrum of machine learning and deep learning models, is rapidly emerging as a prominent research frontier [4, 11–14]. One such study in this area, conducted by Park et al. [5], investigated the impact of cool roofs on energy efficiency in urban areas. Their work posits that the use of cool roofs on urban buildings mitigates the greenhouse effect and enhances energy efficiency, thereby establishing it as a critical research topic. In their study, a roof detection model was developed using Convolutional Neural Network (CNN) based architectures on aerial imagery obtained from South Korea. The detected rooftops were subsequently grouped according to their surface color, and their respective areas were calculated. This methodology determines the potential energy savings achievable through cool roof implementation. Through a cost-benefit analysis, it was concluded that cool roofs significantly enhance urban sustainability.

Assouline et al. [6] performed roof type classification, stating that large-scale photovoltaic (PV) installations on building rooftops represent one of the most efficient and viable options for renewable energy in urban settings. In this study, which utilized a Random Forest classifier, an accuracy of 67% was achieved on a dataset containing building types from Switzerland.

For the accurate identification of rooftop photovoltaic (PV) installations, which is crucial for renewable energy planning and resource assessment, Chaweewat et al. [7] employed an enhanced Mask R-CNN model. Aiming to overcome challenges posed by the complexity of urban environments, diverse roof geometries, and varying illumination conditions, this study used high-resolution satellite imagery as input for the Mask R-CNN. This highly feasible method, used in conjunction with data augmentation techniques to reduce overfitting and improve generalizability, yielded reliable results.

The most recent and comprehensive benchmarking ground in this field is the 2023 IEEE GRSS Data Fusion Competition (DFC23). The contest presented a challenging dataset of 12 fine-grained rooftop classes collected from 17 different cities globally. The winning teams' solutions reflect the most advanced techniques available. The winning team, PIESAT-AI, presented a highly complex system based on the Cascade Mask R-CNN [8] architecture, including domain-adapted pre-training, a dual-backbone structure and a special model ensemble technique called Weighted Segmentation Fusion (WSF). The second-place team IPIU-XDU proposed a three-stage framework combining pre-training, supervised training and semi-supervised learning using a Transformer-based backbone such as BEiTv2-L [9]. Both winning teams used advanced data augmentation and loss function strategies to address the class imbalance problem and resorted to ensemble methods that combine the results of multiple models to achieve the best performance [10].

The pursuit of high accuracy in the literature often leads to more complex models. This study focuses on the applicability of a hybrid architecture that decouples tasks, as an alternative to monolithic systems. Our approach is based on two steps: first, making the classification task more defined by reducing the classes in the DFC23 dataset to four main categories; and second, using the YOLOv11 model, known for its success in the classification

domain. Among the three designed scenarios, our primary hybrid model (MMRotate+YOLOv11-cls) adopts the principle of logical task separation. In this architecture, MMRotate's proficiency in rotated object detection pinpoints the rooftops, while the responsibility of classification is assigned to the YOLOv11 (YOLOv11-cls) architecture. This strategy allows for the utilization of each component's proficiency in its respective domain. Following this logic, an inverse scenario with detection by YOLOv11 and classification by MMRotate was excluded from evaluation, as it is not a suitable method for the targeted classification performance, given that MMRotate is not a classification tool.

2. Material and Method

2.1. Dataset

In this study, the DFC23 Track1 dataset published in 2023 as part of the IEEE GRSS Data Fusion Contest was used as the base dataset. The dataset consists of high-resolution satellite imagery from SuperView-1, Gaofen-2 and Gaofen-3 satellites with different spatial resolutions of 0.5 m, 0.8 m and 1 m. This globally diverse dataset includes images of 17 different cities from 6 continents and 12 different rooftop types.

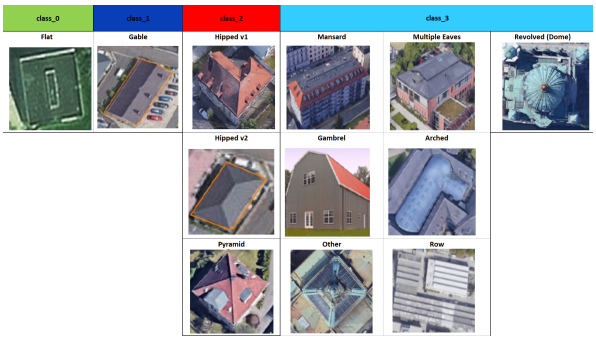


Fig. 1. Rooftop type data set classification.

To simplify the model's learning process and minimize the class imbalance problem in multi-class classification problems, the 12 original rooftop types were divided into four main categories based on the number of labels and morphological similarities, as shown in Figure 1. This regrouping is as follows:

- class_0: Flat roof
- class_1: Gable roof
- class_2: Hipped roof v1, hipped roof v2 and pyramid roof
- class_3: Mansard, gambrel, other, multiple eave, arch, revolved, dome and row roof

The main DFC23 dataset contains 3551 labeled images with a total size of 512×512 pixels. The experimental methodology of this study was designed with a two-stage approach to measure both the basic generalization ability and the field-specific adaptation ability of the models. In the first stage, all models were trained from scratch on the DFC23 training set consisting of 3551 labeled images. To prevent data leakage and ensure that each cluster reflects the class distribution of the main dataset, this splitting process was performed using stratified random sampling with a ratio of 70% training, 20% validation, and 10% testing.

As a second and additional step, fine-tuning was performed to further improve the performance of these baseline models. In this process, the baseline models pre-trained on the DFC23 dataset were fine-tuned using 414 images from the training set provided by Huawei R&D Turkey's customer in China, representing 828 images from the Shanghai region. The remaining 414 images were used as a test dataset for analysis. This specific dataset was selected to meet the real-world requirements of Huawei R&D China's customer and was used to evaluate model performance on Shanghai's unique urban architecture. The purpose of this fine-tuning process was to improve model performance by adapting the models' general knowledge to domain specific architectural features. This two-stage approach aims to measure the fundamental generalization ability of the models and quantitatively demonstrate how much this domain specific finetuning improves performance.

2.2. Method

Rooftop type detection and classification from satellite imagery is an actively researched topic today, requiring the application of advanced image processing and deep learning techniques. In this study, two different and powerful deep learning architectures, MMRotate [16] and YOLOv11 [17], are selected for rooftop type detection. This choice is based on their task-specific performance, which has been proven in the literature. MMRotate is a standard-setting framework designed for the detection of objects with slopes and different orientations, which are frequently encountered in satellite imagery, and is particularly good at precise positioning of bounding boxes. YOLOv11, on the other hand, is recognized as one of the latest and most powerful architectures for real-time detection applications, thanks to its balance between speed and accuracy.

Before model training, a critical data format conversion process was applied to make the dataset compatible with the input requirements of each architecture. In this context, labels in COCO format were converted to DOTA format for the MMRotate framework, which is specialized for rotated object detection, and to the standard YOLO format for the general-purpose detection model YOLOv11.

In this study, three main scenarios representing different architectural philosophies are designed and compared, as shown in Figure 2:

1. End-to-End Rotated Object Detection Framework: In the first scenario, the MMRotate model, which has demonstrated high performance in detecting tilted objects in satellite images, was employed. In this approach, the model performs object localization, bounding box regression, and classification tasks in an integrated, end-to-end structure.

2. Decoupled Hybrid Detection-Classification Pipeline: The second scenario presents a two-stage pipeline architecture in which the detection and classification tasks are logically separated. The basic logic of this approach is to delegate each task to the expert most capable of performing that task:

2.1. Detection Stage: In the first stage, the MMRotate model generates object proposals to determine the locations and orientations of roofs with high accuracy.

2.2. Classification Stage: In the second stage, a YOLOv11 classification task model trained solely for classification is used. This classifier was trained on a special dataset created by cropping objects from the ground truth data and resizing them to 224×224 pixels, using the proven ResNet-18 backbone architecture. It is important to center each bounding box using the building's height information during this cropping process, as shown in Figure 3. This ensures that the entire rooftop structure is captured and the classifier can take advantage of the distinctive features of the roofs

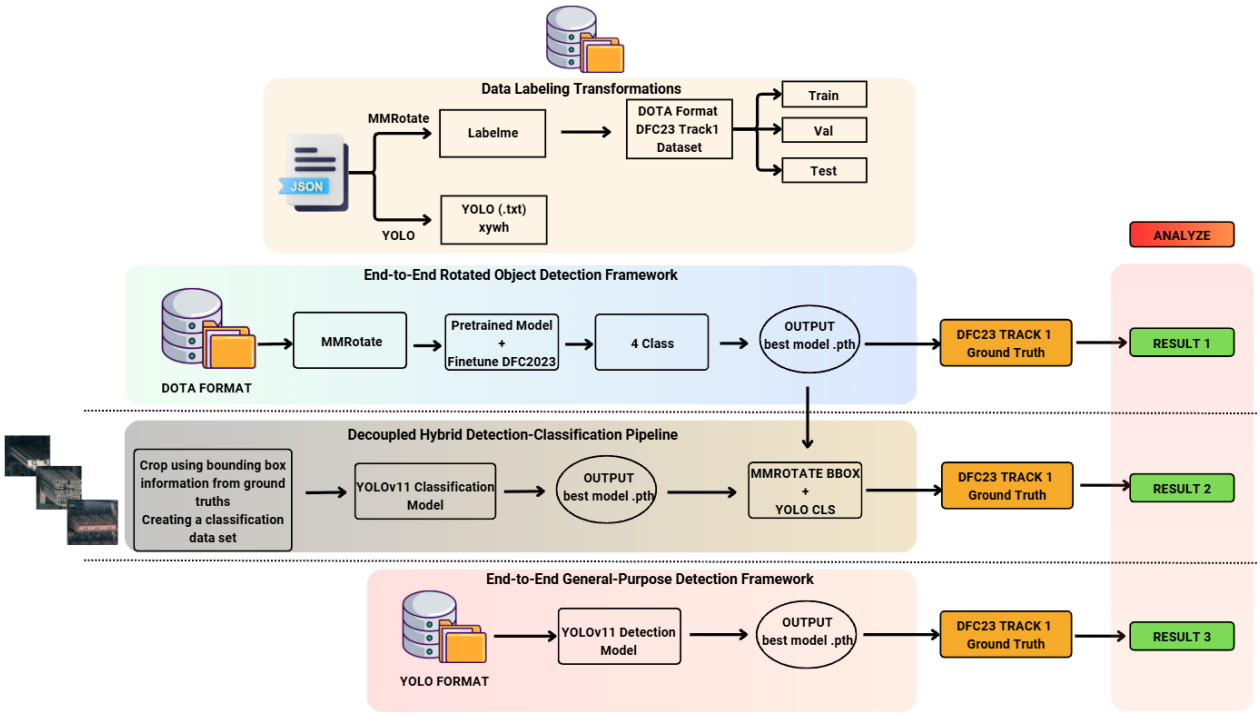
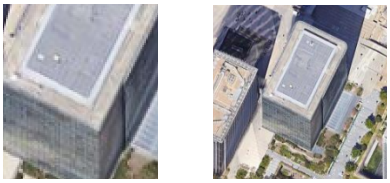


Fig. 2. Data set transformations and flowchart summary for rooftop type detection.

for better generalization. During inference, the bounding box proposals generated by MMRotate are sent to this specific classifier to assign the final rooftop type label.



(a) (b)

Fig. 3. Illustration of the height-centered cropping process. The naive crop in (a) can exclude critical roof features, while the proposed method in (b) ensures the entire roof is captured

3. End-to-End General-Purpose Detection Framework: In the third scenario, the YOLOv11 detection task model, known for its effective balance between speed and accuracy, was trained as a standard end-to-end model that performs detection and classification tasks simultaneously. This scenario serves as a reference point for evaluating the performance of a modern and efficient general-purpose detection model.

3. Experiments and Conclusions

This section details the experimental setup, presents the quantitative and qualitative results obtained from the three proposed scenarios, and provides an analysis of the models' comparative performance.

To ensure that all experimental scenarios can be compared fairly, the training processes have been standardized. The models were trained on an Nvidia Tesla V100 GPU with a batch size of 16, an initial learning rate of 0.001, and a total of 100 epochs. ADAM was used as the optimization algorithm. The performance of the models was evaluated using the Intersection over Union (IoU) metric, which measures the intersection area between the predicted bounding boxes and the ground truth boxes. An IoU threshold value of 0.5 was set for a detection to be considered correct. The final results were analyzed using the Precision, Recall, and F1-Score metrics.

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{1}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{2}$$

$$F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{3}$$

Figure 4a contains the actual reference information for different classes for the selected sample image. Figures 4b, 4c, 4d represent the rooftop detection and classification results of the MMRotate, YOLOv11-det, MMRotate + YOLOv11-cls models, respectively, on the same image. When these results are compared with the ground truth information in Figure 4a, it is observed that the closest results are obtained with MMRotate + YOLOv11-cls(4d).

The performance of the three architectural scenarios was evaluated on the DFC23 general test set, and the model with the highest performance, MMRotate+YOLOv11-cls, was also evaluated on the Shanghai test set. The results, including key performance metrics are summarized in Table 1 and Table 2.

Table 1 compares three architectural scenarios on the baseline DFC23 dataset. According to these results, the hybrid model achieved the highest overall F1-Score of 0.85. This score is higher than that of the standalone MMRotate model (0.81 F1-Score) and the end-to-end YOLOv11 detection model (0.71 F1-Score). The

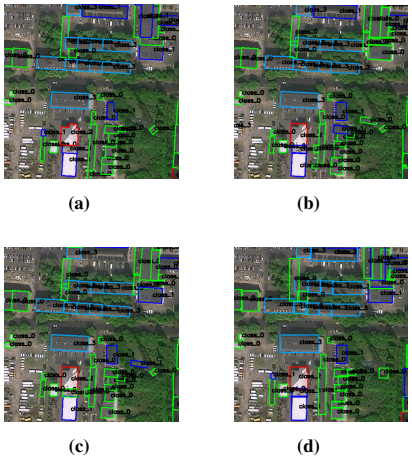
Table 1. YOLOv11 and MMRotate Performance Results on the Baseline Dataset.

Class	MMRotate Model			YOLOv11-det Model			Hybrid Model*		
	Precision	Recall	F1-Score	Precision	Recall	F1-Score	Precision	Recall	F1-Score
class_0	0.82	0.76	0.79	0.67	0.73	0.70	0.84	0.80	0.82
class_1	0.84	0.81	0.82	0.74	0.70	0.72	0.86	0.84	0.85
class_2	0.85	0.83	0.84	0.77	0.75	0.76	0.88	0.85	0.87
class_3	0.83	0.78	0.80	0.72	0.62	0.67	0.87	0.83	0.85
Overall	**0.84**	**0.80**	**0.81**	**0.73**	**0.70**	**0.71**	**0.86**	**0.83**	**0.85**

*Hybrid Model refers to the MMRotate + YOLOv11-cls architecture.

Table 2. Performance Comparison of the Hybrid Model: Base vs. Shanghai Fine-tuned.

Class	Base Hybrid Model			Shanghai Fine-tuned Hybrid Model		
	Precision	Recall	F1-Score	Precision	Recall	F1-Score
class_0	0.84	0.80	0.82	0.87	0.84	0.86
class_1	0.86	0.84	0.85	0.90	0.86	0.88
class_2	0.88	0.85	0.87	0.91	0.93	0.92
class_3	0.87	0.83	0.85	0.90	0.88	0.89
Overall	**0.86**	**0.83**	**0.85**	**0.90**	**0.88**	**0.89**



(a) **(b)** **(c)** **(d)**

Fig. 4. Comparative results on sample images: a) Ground Truth; b) MMRotate; c) YOLOv11-det; d) MMRotate + YOLOv11-cls

hybrid model also recorded the highest F1-Score for each of the four separate rooftop classes. This result suggests that separating the detection and classification tasks may be a useful approach for this problem.

Table 2 shows the results of the hybrid model before and after fine-tuning on the Shanghai dataset to evaluate the effect of domain-specific adaptation. The fine-tuning process resulted in improvements across all reported performance metrics. The overall F1-Score increased from 0.85 to 0.89, while Precision increased to 0.90 and Recall to 0.88. An increase in F1-Score was observed for each class, with Class 2 reaching a score of 0.92. In addition, the computational efficiency of the fine-tuned hybrid model was evaluated on the Shanghai test set to assess its suitability for practical applications.In terms of computational efficiency, the hybrid model has an average inference time of 6.62 ms per image (measured on a Tesla V100 GPU). This processing time is only slightly slower than the standalone YOLOv11-det (5.30 ms) and MMRotate (6.00 ms) models. What matters is that this small increase in latency is combined with a significant improvement in accuracy. The hybrid model achieved an overall F1 score of 0.85, outperforming both the base YOLOv11-det (0.71 F1 score) and MMRotate (0.81 F1 score) models. This result highlights that the decomposed architecture provides a highly suitable speed-accuracy trade-off for this task.

4. Discussions

An important finding from the experiments is the performance of the hybrid MMRotate+YOLOv11-cls model, which achieved the highest average F1-Score. This result shows that dividing the problem into specific tasks provides useful synergy. The combination of MMRotate's rotated object localization and YOLOv11's classification backbone emerged as an effective strategy in our tests. Additionally, the robust performance of the standalone YOLOv11 model is worth mentioning, as it demonstrates the ability of modern general-purpose detectors to perform fine-grained classification tasks on satellite imagery. The study also examined the model's adaptability through domain-specific fine-tuning. When comparing the results of the base model with those of the fine-tuned model on the Shanghai subset, a performance improvement was observed. This demonstrates that the proposed architecture can be adapted to a new geographic region with unique architectural characteristics using a relatively small dataset. This finding suggests potential for practical applications where models need to be tailored to specific target areas. It is difficult to make a direct numerical comparison with the winners of the DFC23 competition due to differences in task definitions (object detection and semantic segmentation) and primary metrics (F1-Score and mAP). However, a conceptual comparison can be illuminating. The winning solutions typically relied on highly complex systems, such as clusters of large models, to achieve maximum accuracy, which can be associated with significant computational costs. Our work explores a different architectural philosophy. The winning strategies can be seen as a kind of **complexity combination** where multiple models performing the same task are combined. In contrast, our approach is based on **task-specific specialization**. This is not a community but a modular process that divides the problem into two separate subtasks:

1. **Localization**: Answering the question "Where is the rooftop?" using MMRotate.
2. **Classification**: Then, using YOLOv11 to answer the question "What type of rooftop is this?"

This architectural choice represents a functional and modular approach. By delegating different tasks to specialized models, the resulting pipeline offers a potentially more interpretable and flexible structure, which can be a practical alternative to monolithic, highly complex models for certain remote sensing challenges. Building upon the successful domain adaptation demonstrated with the Shanghai dataset, future work will focus on developing a highly generalized model for the broader China region. The next steps will involve extending the fine-tuning process to a more comprehensive dataset that includes the diverse architectural styles found across various Chinese cities, aiming to optimize the model's performance and robustness for regional applications.

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